FLAT FINITE PRESENTATION MORPHISMS OF LIMITS ARE CONSTANT ON THE SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5432-5459, following the statement at 5423-5430.

Editorial note. No omissions; the redefinition of the system via the remark is explicit.

Allowed standard prerequisites (editorial).

- Morphisms of limits over good diagrams. lemma-morphism-good-diagram (same chapter, card L26309, cited at ~5425). Notation, i_0, and the system morphisms f_i.
- Descent of finite presentation morphisms. lemma-descend-finite-presentation (same chapter, card L26286, cited at ~5433-5436). fp model U -> Y_i with X = Y x_{Y_i} U.
- Descent of flatness. lemma-descend-flat (same chapter, cited at ~5437-5438). U -> Y_i flat after increasing i.
- Well-definedness of the redefined system. remark-finite-type-gives-well-defined-system (same chapter, cited at ~5439-5443). Replaces the defining diagram by X -> U -> B_i.
- Scheme theoretic image under flat base change. Morphisms of Spaces, Lemma spaces-morphisms-lemma-flat-base-change-scheme-theoretic-image (cross-chapter, cited at ~5446-5451). The scheme theoretic image of X -> U is U.

Source reference. Original lemma 07SK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2148–2148. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08K0: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1969–1969. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CPA: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5324–5324. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark 0CPB: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5404–5404. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 089E: focused-seeds/geometry/S9-spaces-limits-morphism-good-diagram-flat/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Lemma 0CPA. If f is flat and of finite presentation, then there exists an $i_3 > i_0$ such that for $i \geq i_3$ we have f_i is flat, $X_i = Y_i \times_{Y_{i_3}} X_{i_3}$, and $X = Y \times_{Y_{i_3}} X_{i_3}$.*

Proof. By Lemma 07SK we can choose an $i \geq i_2$ and a morphism $U \to Y_i$ of finite presentation such that $X = Y \times_{Y_i} U$ (this is where we use that f is of finite presentation). After increasing i we may assume that $U \to Y_i$ is flat, see Lemma 08K0. As discussed in Remark 0CPB we may and do replace the initial diagram used to define the system $(X_i)_{i \geq i_1}$ by the system corresponding to $X \to U \to B_i$. Thus $X_{i'}$ for $i' \geq i$ is defined as the scheme theoretic image of $X \to B_{i'} \times_{B_i} U$.

Because $U \to Y_i$ is flat (this is where we use that f is flat), because $X = Y \times_{Y_i} U$, and because the scheme theoretic image of $Y \to Y_i$ is Y_i, we see that the scheme theoretic image of $X \to U$ is U (Morphisms of Spaces, Lemma 089E). Observe that $Y_{i'} \to B_{i'} \times_{B_i} Y_i$ is a closed immersion for $i' \geq i$ by construction of the system of Y_j. Then the same argument as above shows that the scheme theoretic image of $X \to B_{i'} \times_{B_i} U$ is equal to the closed subspace $Y_{i'} \times_{Y_i} U$. Thus we see that $X_{i'} = Y_{i'} \times_{Y_i} U$ for all $i' \geq i$ and hence the lemma holds with $i_3 = i$. $\square$